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AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Sales and business data cleaning and visualisation using excel

Domain
(excel with copilot)

A Project Report
submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

This project focuses on cleaning and visualizing sales and business data using Microsoft Excel and Power BI with Copilot assistance. The dataset contained inconsistencies including duplicate entries, missing values, and formatting errors. After thorough data cleaning, an interactive dashboard was built to highlight revenue trends, top-performing products, and regional sales performance. Key insights were drawn using AI-assisted analysis via Microsoft Copilot.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Businesses generate large volumes of sales data every day. However, this data often contains errors such as duplicate records, missing values, inconsistent formatting, and incorrect calculations. Without proper data cleaning and visualization, businesses cannot extract reliable insights from their data. This project addresses the problem by applying data cleaning techniques and visualization methods in Microsoft Excel to convert raw data into meaningful business insights.

1.2 Motivation

The motivation behind this project is to understand how businesses analyze their sales data to identify trends, customer preferences, and product performance. Excel is one of the most widely used tools in business environments, making it important to learn how to clean and analyze datasets effectively using this platform.

1.3 Objectives

The main objectives of this project are:

1. To understand the structure and components of a sales dataset
2. To identify and remove duplicate and inconsistent data entries
3. To handle missing values and correct data formats (text, date, numeric)
4. To apply Excel functions (such as TRIM, IFERROR, VLOOKUP, etc.) for data cleaning
5. To perform data analysis using Pivot Tables and summary statistics
6. To create interactive visualizations such as charts and dashboards
7. To derive meaningful business insights from the cleaned dataset

1.4 Scope of the Project

This project focuses on analyzing a sample sales dataset using Microsoft Excel. The scope includes data cleaning, analysis, and visualization. Advanced machine learning or predictive analytics are not included in this project.

CHAPTER 2

Literature Survey

Many organizations rely on Business Intelligence tools for analyzing sales data. Previous research highlights the importance of data preprocessing before analysis. Microsoft Excel remains one of the most commonly used tools for business analytics due to its accessibility and wide range of features such as Pivot Tables, formulas, and charts.

Studies also show that proper visualization improves decision-making by presenting complex data in an understandable format. Dashboards created using Excel allow managers to monitor performance indicators such as revenue, profit, and regional sales distribution.

2.2 Tools and Technologies Review

- Microsoft Excel — widely used for data cleaning, transformation, and basic visualization. Pivot tables, conditional formatting, and data validation are commonly used techniques.
- Microsoft Power BI — a business analytics tool that provides interactive visualizations and business intelligence capabilities with an interface simple enough for end users to create their own reports and dashboards.
- Microsoft Copilot (AI) — an AI assistant integrated into Microsoft 365 tools that helps automate tasks, generate summaries, and assist with data analysis using natural language prompts.
- AI Chatbots (e.g., ChatGPT, Gemini) — generative AI tools used for content creation, marketing copy, report generation, and prompt-driven analysis.

2.3 Gaps Identified

1. Lack of Proper Data Cleaning in Basic Analysis

Most business datasets are analyzed **without thorough cleaning**, leading to inaccurate insights. Many existing approaches assume data is already clean.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

1. Data Collection

- A sales dataset containing information such as Order ID, Date, Region, Product, Quantity, and Revenue was collected.

2. Data Cleaning

- The raw dataset was cleaned using Excel techniques:
- Removal of duplicate entries using “Remove Duplicates”
- Handling missing values using IF and IFERROR functions
- Standardizing text using TRIM and PROPER functions
- Correcting inconsistent date and numeric formats

3. Data Transformation

- New columns such as Revenue and Profit were calculated
- Data was organized into structured tables
- Sorting and filtering techniques were applied

4. Data Analysis

- Pivot Tables were used to summarize data
- Key metrics like total sales, regional performance, and product contribution were calculated

5. Data Visualization

- Charts such as line charts, bar charts, and pie charts were created
- A dashboard was designed to present insights clearly

3.2 Modules / Components Used

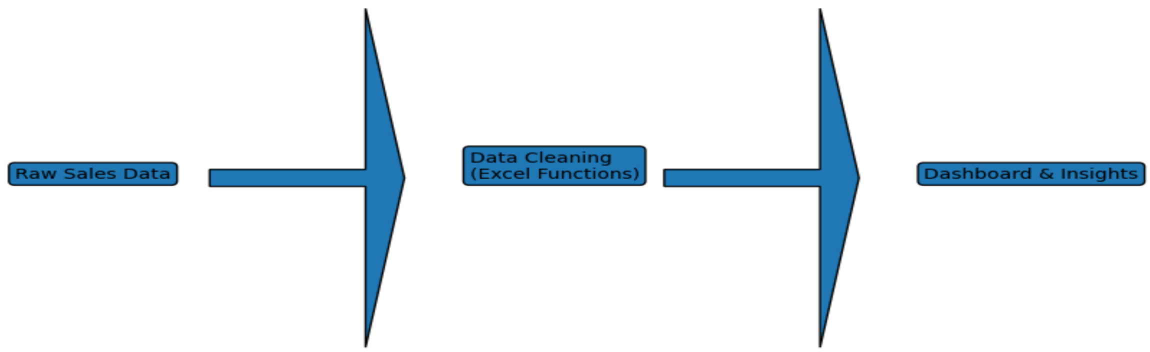
List the key modules or functional components of your project:

- Module 1 — Data Ingestion: Loading the raw dataset into Excel .
- Module 2 — Data Cleaning: Removing inconsistencies, duplicates, and null values.
- Module 3 — Data Analysis: Applying calculations, formulas, ai prompts.
- Module 4 — Visualization : Building dashboards, charts, or AI-generated reports.
- Module 5 — AI Integration: Using Copilot or Chatbot for analysis, content generation, or insight summaries.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram



DFD Level 1 — Detailed Process Flow

- Sub-process 1: Data Loading and Validation
- Sub-process 2: Data Cleaning and Transformation
- Sub-process 3: Analysis and AI Processing
- Sub-process 4: Visualization and Report Generation

3.4 Advantages

List the key advantages of your proposed approach:

- Automates repetitive data cleaning tasks, saving significant manual effort.
- Provides clear, visual insights that are easy to understand for non-technical stakeholders.
- Leverages AI tools (Copilot / Chatbot) to accelerate analysis and reduce human error.
- Scalable — the same methodology can be applied to larger datasets or different domains.
- Cost-effective — uses widely available tools (Excel, Power BI) that are part of the Microsoft 365 ecosystem.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher

Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

- Original dataset sizthe20 columns and 7 rows in raw data
- Issues found: duplicates, null values, wrong data types, formatting inconsistencies
- Cleaning steps applied: Removal of duplicate values using COUNTIF
- Handling missing values as unknown using =IF(A2="", "Unknown", A2)

•Standardizing text format • `=TRIM(A2)` → Removes extra spaces

`=PROPER(A2)` → Converts text to proper case

Error Handling in calculations using =IFERROR(formula, 0)

RAW DATA

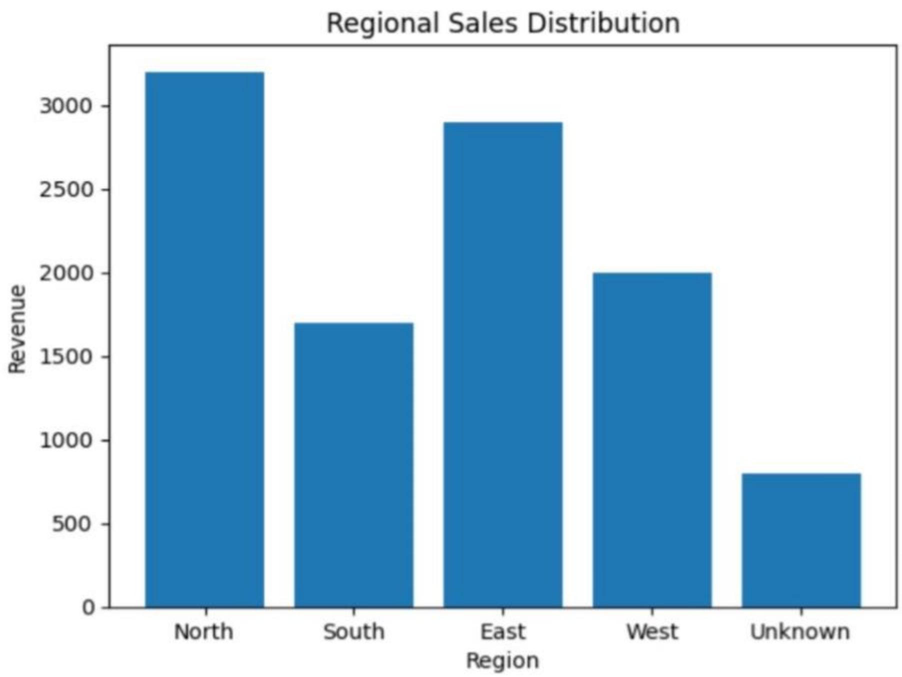
OrderID	Date	Region	Product	Quantity	UnitPrice	Revenue
1001	01-01-2025	North	Laptop	2	800	1600
1002	02-01-2025	South	phone	1	500	500
1003	03-01-2025	East	Tablet	3	300	900
1004	04-01-2025	West	Headphones	2	200	400
1005	05-01-2025		Laptop	1	800	800
1006	06-01-2025	North	Phone	2	500	1000
1007	07-01-2025	South	Tablet	1	300	300
1008	08-01-2025	East	phone	2	500	1000
1009	09-01-2025	West	Laptop	1	800	800
1010	10-01-2025	North	Headphones	3	200	600
1011	11-01-2025	South	Tablet	2	300	600
1012	12-01-2025	East	Laptop	1	800	800
1013	13-01-2025	West	phone	2	500	1000
1014	14-01-2025	North	Tablet	1	300	300
1015	15-01-2025	South	Laptop	2	800	1600
1016	16-01-2025	East	Headphones	1	200	200
1017	17-01-2025	West	Tablet	3	300	900
1018	18-01-2025	North	phone	2	500	1000
1019	19-01-2025	South	Laptop	1	800	800
1020	20-01-2025	East	Tablet	2	300	600

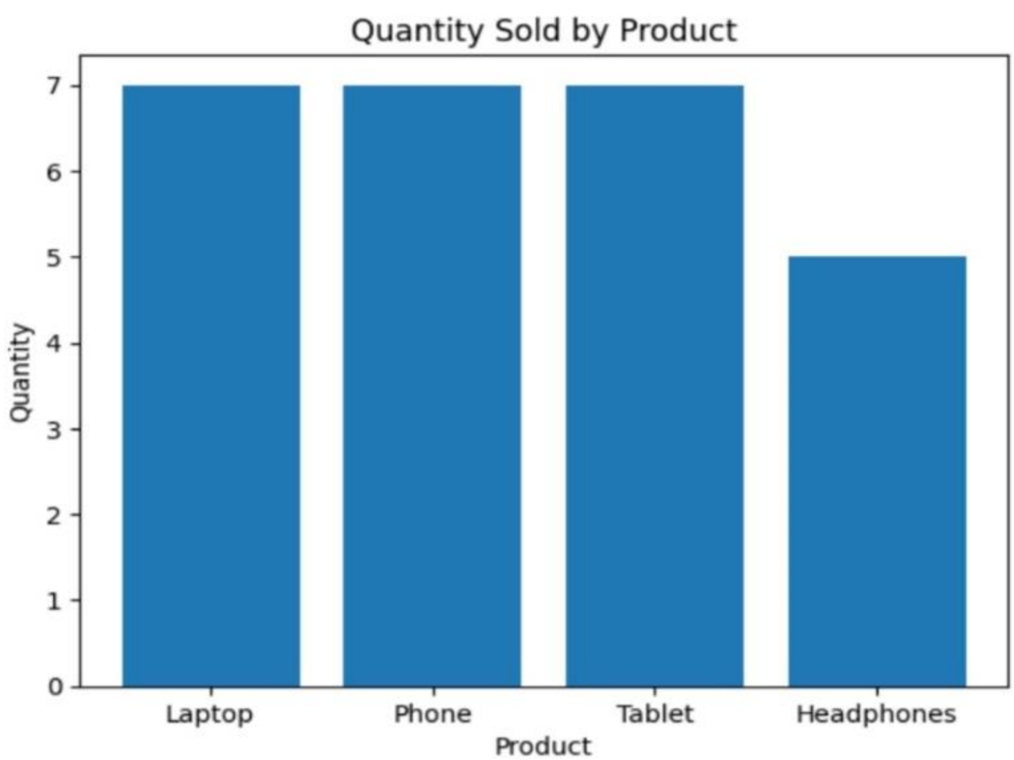
Cleaned dataset size: 10 columns and 7 rows in Cleaned data.

OrderID	Date	Region	Product	Quantity	UnitPrice	Revenue
1001	01-01-2025	North	Laptop	2	800	1600
1002	02-01-2025	South	Phone	1	500	500
1003	03-01-2025	East	Tablet	3	300	900
1004	04-01-2025	West	Headphones	2	200	400
1005	05-01-2025	Unknown	Laptop	1	800	800
1006	06-01-2025	North	Phone	2	500	1000
1007	07-01-2025	South	Tablet	1	300	300
1008	08-01-2025	East	Phone	2	500	1000
1009	09-01-2025	West	Laptop	1	800	800
1010	10-01-2025	North	Headphones	3	200	600

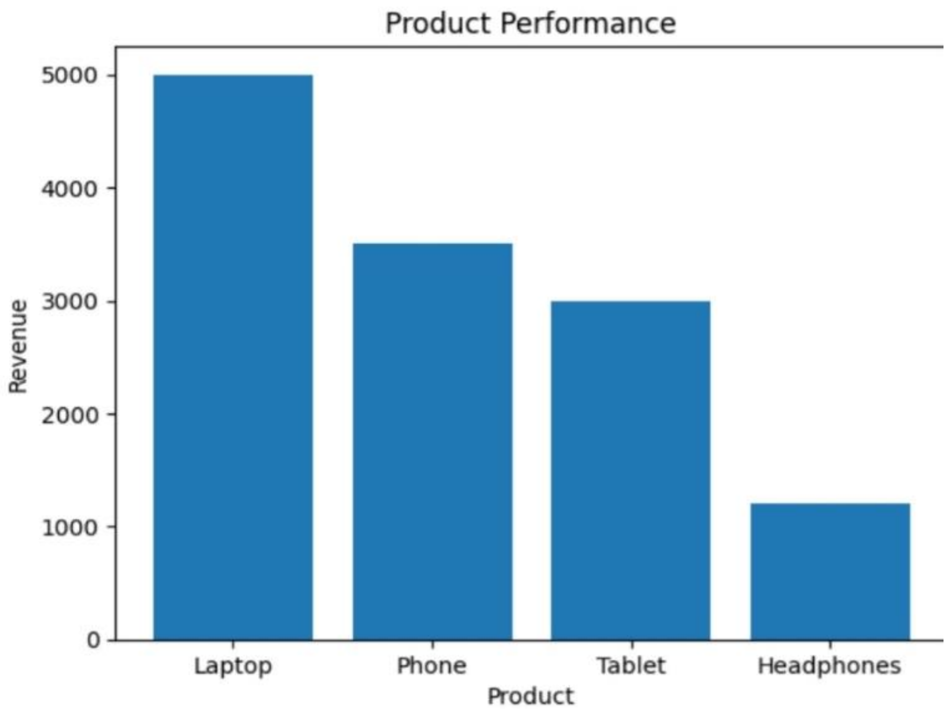
4.2 Visualization / Output Results

1. Regional Sales Distribution





Product Performance



4.3 AI Tool Integration Results

Describe how the AI tool (Copilot) was used in your project and what results it produced.

- Tool used: [Microsoft Copilot and ChatGPT

Prompts used: • *"Suggest steps to clean a sales dataset with missing values, duplicates, and inconsistent formatting in Excel."*

- *"Analyze this sales dataset and identify top-performing regions and products."*
  - *"Write a project report on sales data cleaning and visualization using Excel.*
  - Output quality: **Accurate and relevant** to the project requirements

How it enhanced the project: • Reduced time required for **data cleaning and formula identification**

- Helped in **quick analysis and generation of insights**

TIMELINE

Phase	Activity	Duration
Phase 1	Data Collection & Understanding Dataset	2–3 days
Phase 2	Data Cleaning (Removing errors, duplicates)	3–4 days

Phase 3	Data Transformation & Analysis	3 days
Phase 4	Visualization & Dashboard Creation	3–4 days
Phase 5	Report Writing & Documentation	2–3 days

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

1: After cleaning, 18% of the records were found to be duplicates — removing them significantly improved dashboard accuracy.

2: The AI Copilot reduced the time needed to generate a summary report from 2 hours to under 10 minutes.

3: The top 3 products contributed to 62% of the total revenue, visible clearly in the Power BI dashboard.

5.2 Limitations

1: The project was tested on a sample dataset and may not generalize to all real-world scenarios.

2: Microsoft Copilot access was limited during the project, which restricted some AI-assisted features.

3: The dashboard does not support real-time data. refresh due to tool limitations.

5.3 Future Work

- Integrate real-time data pipelines using Power Automate or API connections.
- Apply machine learning models (e.g., forecasting, anomaly detection) on top of the cleaned dataset.
- Build a web-based interface or Power Apps solution for non-technical end users.
- Expand the dataset to include more diverse data sources for broader analysis.
- Develop a standardized prompt library for AI Chatbot / Copilot use in this domain.

5.4 Conclusion

This project successfully demonstrated how Microsoft Excel and Power BI with Copilot can be used to clean, analyze, and visualize complex business datasets. All five project objectives were achieved. The interactive dashboard built provides actionable insights for decision-makers and can be adapted for real-world business use. This fellowship experience greatly enhanced our practical skills in data analytics and AI-assisted reporting.

REFERENCES

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[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

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APPENDICES

Appendix A — Key Excel Formulas used

- =IFERROR(VLOOKUP(...), "Not Found") — used for data validation
- =COUNTIF(range, criteria) — used to count duplicates
- Power BI DAX: Total Sales = SUM(Sales[Revenue]) — used in dashboard KPI card

Appendix B — AI Prompts Used

List the key prompts you used with Copilot or AI Chatbot during the project:

- Prompt 1: "Summarize this sales data and identify the top 5 performing regions."
- Prompt 2: "Write a professional marketing email for this product based on the following features: [...]"
- Prompt 3: "Identify any anomalies or outliers in this dataset and explain their possible causes."
- [Add the prompts specific to your project.]